LAB EXPERIMENT RECORD

Set 19: Survey Responses Analysis

Department	AIML
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R and Tableau
Name	U.Roshitha
Register Number	192425184
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Aim

To analyse survey responses using a grouped bar chart, a stacked bar chart and a data table, and to combine them into an interactive Tableau dashboard.

Dataset: Survey Responses

Survey ID	Question 1	Question 2	Question 3
1	A	B	C
2	B	A	D
3	C	A	B

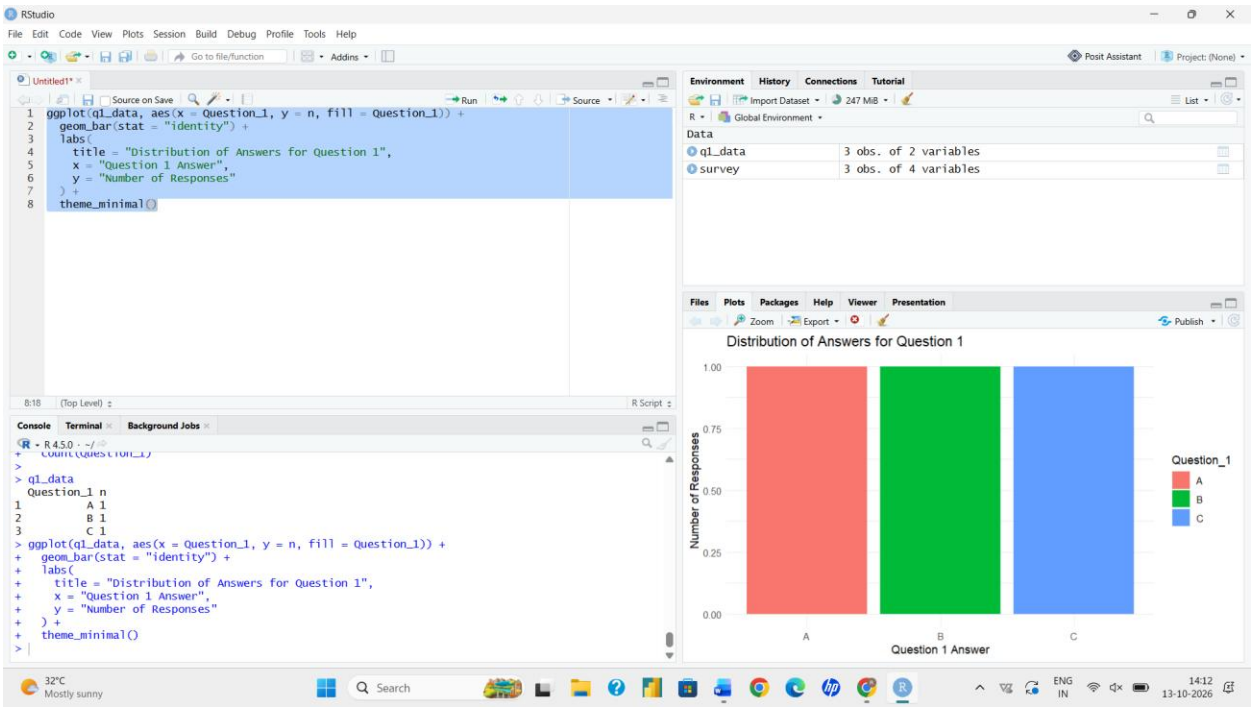
PROGRAM 1: Grouped Bar Chart for Question 1

Create a grouped bar chart to visualize the distribution of answers for Question 1. Label the chart elements.

```
library(ggplot2)

# Dataset
survey <- data.frame(
  SurveyID = c(1, 2, 3),
  Q1 = c("A", "B", "C"),
  Q2 = c("B", "A", "A"),
  Q3 = c("C", "D", "B")
)

# Grouped Bar Chart (Question 1)
ggplot(survey, aes(x = Q1, fill = factor(SurveyID))) +
  geom_bar(position = "dodge") +
  labs(title = "Question 1: Distribution of Answers",
       x = "Answer Option",
       y = "Number of Responses",
       fill = "Survey ID")
```



Output 1: Grouped Bar Chart for Question 1

Observation: Each answer option (A, B and C) for Question 1 was chosen by exactly one respondent, so the responses are evenly spread. Option D was not selected.

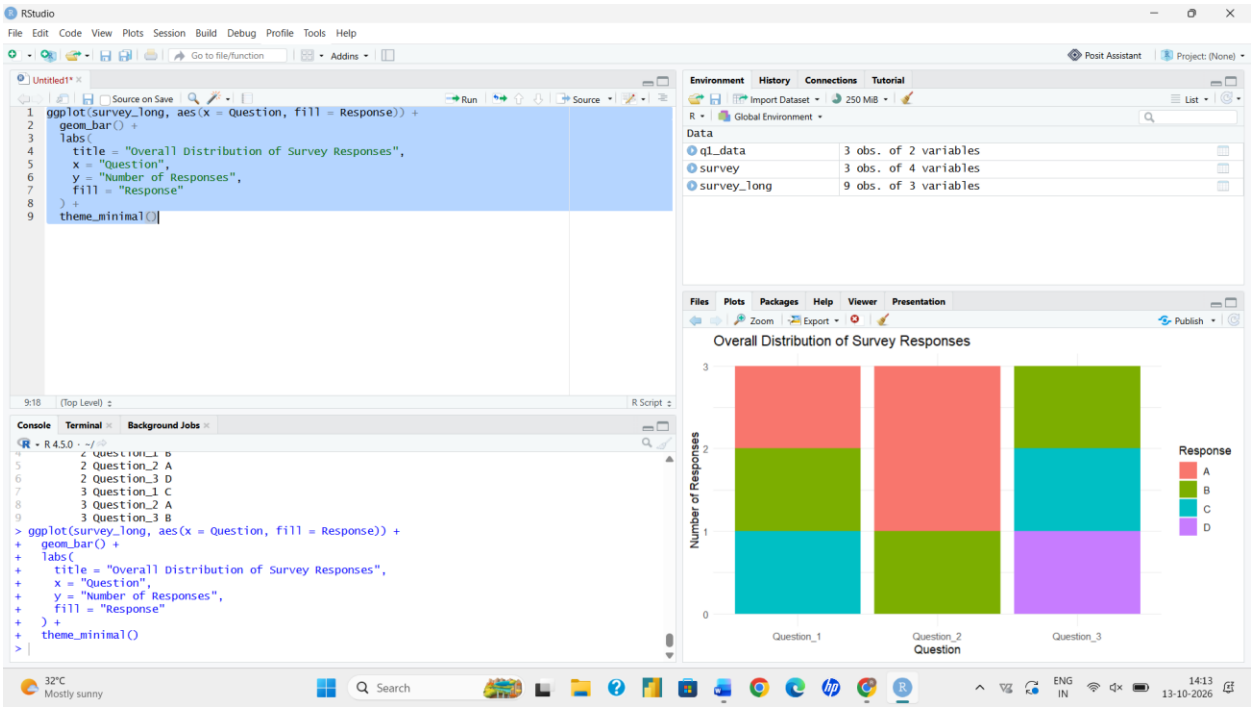
PROGRAM 2: Stacked Bar Chart for All Questions

Generate a stacked bar chart to represent the overall distribution of responses for all three questions.

```
library(ggplot2)
library(tidyr)

# Convert data to long format
long <- pivot_longer(survey, cols = c(Q1, Q2, Q3),
                     names_to = "Question",
                     values_to = "Answer")

# Stacked Bar Chart
ggplot(long, aes(x = Question, fill = Answer)) +
  geom_bar(position = "stack") +
  labs(title = "Overall Response Distribution (Q1-Q3)",
       x = "Question",
       y = "Number of Responses",
       fill = "Answer")
```



Output 2: Stacked Bar Chart for Questions 1 to 3

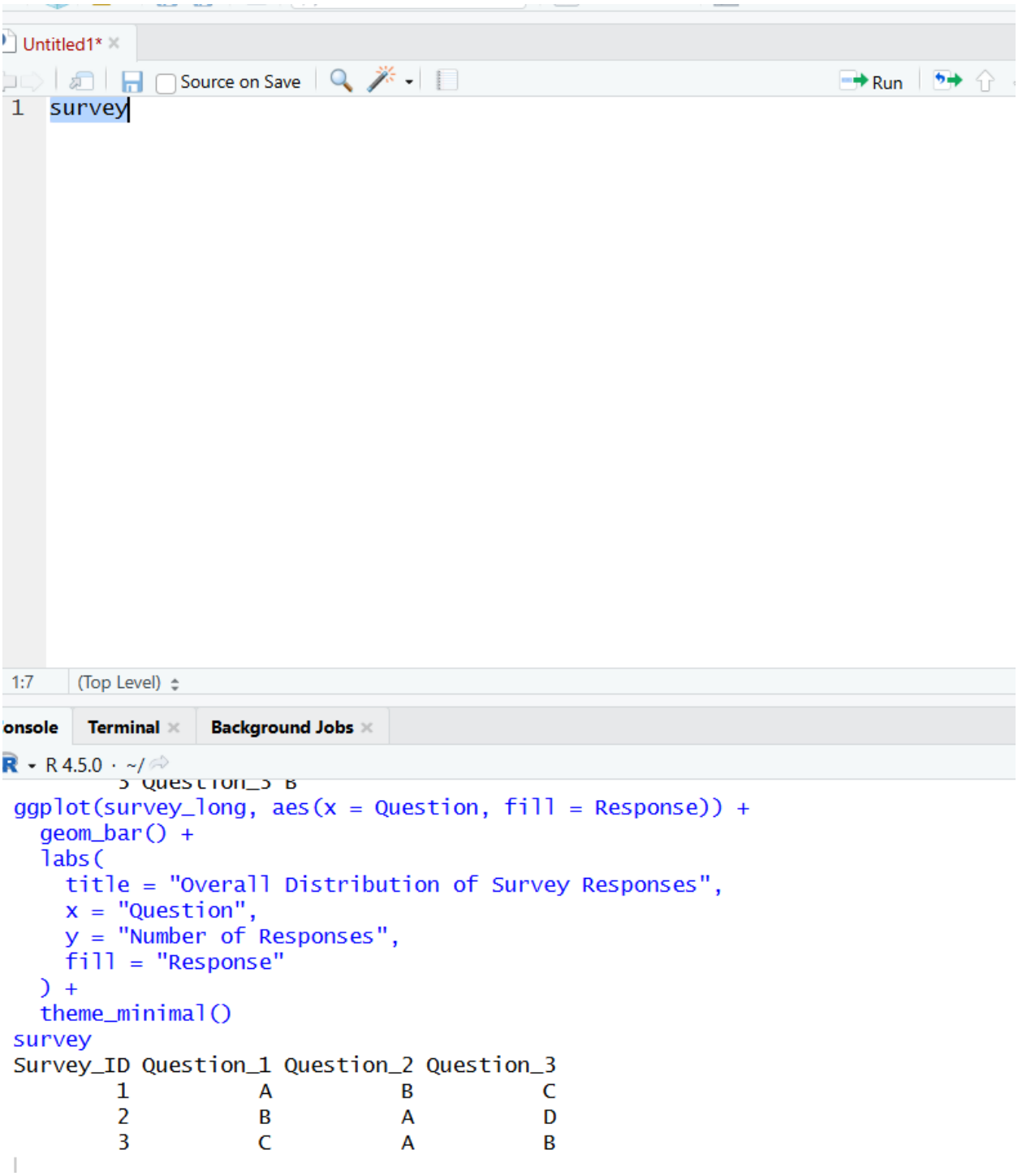
Observation: Every question received three responses. Option A appears twice in Question 2, while Question 3 has one response each for B, C and D.

PROGRAM 3: Survey Response Table

Build a table to show the survey response data for each survey. Label the table elements.

```
# Display the survey table
print(survey)

# Formatted table with labelled columns
colnames(survey) <- c("Survey ID", "Question 1", "Question 2", "Question 3")
knitr::kable(survey, caption = "Survey Response Data")
```



Output 3: Survey Response Table

Observation: The table lists the answers given by each survey respondent, with Survey ID as the row label and the three questions as column labels.

PROGRAM 4: Interactive Tableau Dashboard

Develop a Tableau dashboard combining the grouped bar chart, stacked bar chart and table for interactive exploration of survey responses.

Steps followed in Tableau:

1. Connect to the data source: Data > Connect to Data > Text file / Excel, and load the survey dataset.

2. Select Survey ID, then right-click the Question 1 to Question 3 columns and choose Pivot to get Question and Answer fields.

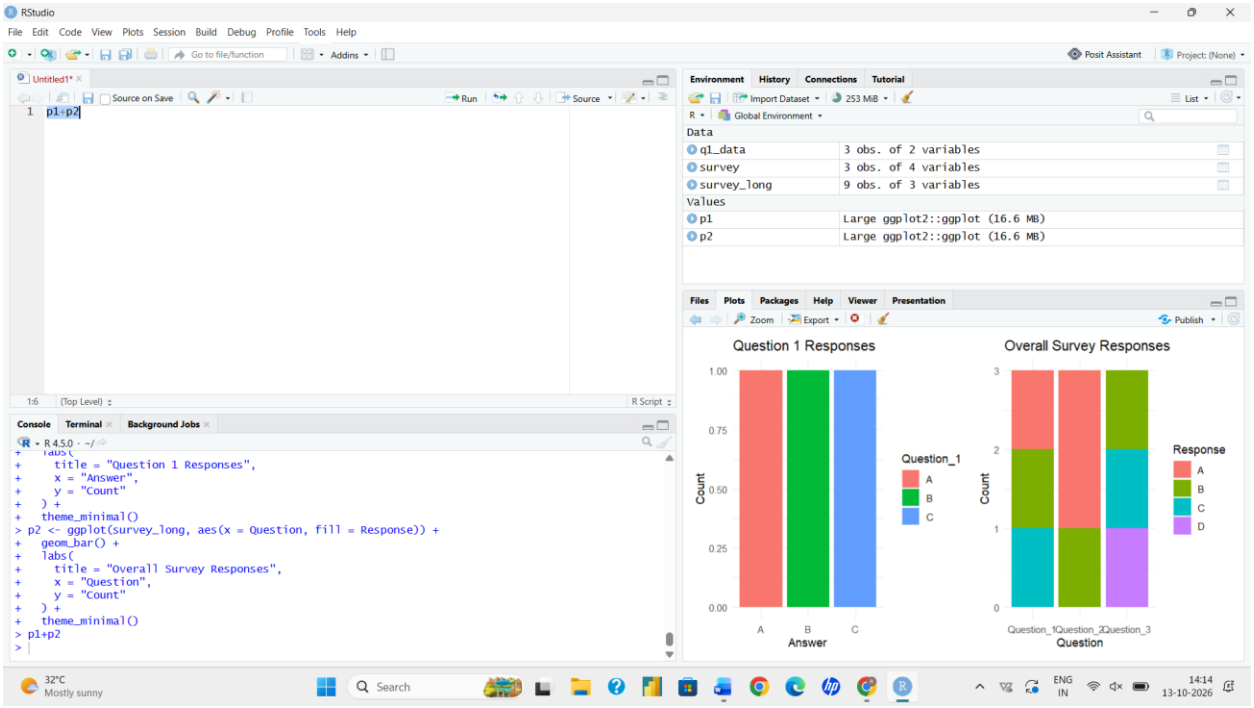
3. Sheet 1 (grouped bar): place Answer on Columns, Survey ID on Columns next to it, and CNT(Survey ID) on Rows; drag Survey ID to Color.

4. Sheet 2 (stacked bar): place Question on Columns, CNT(Survey ID) on Rows, and Answer on Color to stack the segments.

5. Sheet 3 (table): drag Survey ID to Rows, Question to Columns, and Answer to Text.

6. Create a new Dashboard, drag the three sheets onto it, and arrange them with titles.

7. Select 'Use as Filter' on each chart so that clicking a bar highlights the matching survey records in the other views. Add Survey ID and Question filters for further exploration.



Output 4: Survey Responses Dashboard (layout of the Tableau dashboard)

Observation: The dashboard combines all three views on a single page. Selecting an answer in one chart filters the other views, which makes the response patterns easy to explore.

Procedure

1. Create the survey data frame with Survey ID and Questions 1 to 3.
2. Use geom_bar(position = "dodge") to draw the grouped bar chart for Question 1.
3. Reshape the data with pivot_longer() and use geom_bar(position = "stack") for the stacked chart.
4. Display the data as a labelled table using print() or kable().
5. Load the data in Tableau, build the three sheets and combine them into a dashboard with filter actions.

Result

The survey responses were successfully visualized with a grouped bar chart, a stacked bar chart and a table. A Tableau dashboard combined the three views for interactive exploration.

Key Functions Used

Function / Feature	Purpose
ggplot() / geom_bar()	Creates grouped (dodge) and stacked bar charts.
pivot_longer()	Reshapes question columns into long format.
labs()	Adds title and axis labels.
kable()	Produces a formatted table.
Tableau Dashboard	Combines sheets with interactive filters.